|INCH-POUND|

MIL-C-17/193B
20 February 1991
SUPERSEDING
MIL-C-17/193A(EC)
10 August 1987

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/193-00001 UNARMORED, M17/193-00002 ARMORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-17.

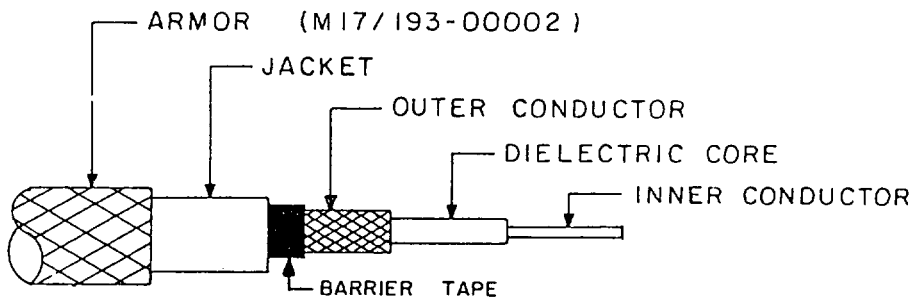


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details		
Inner conductor	Solid bare copper wire. Diameter: .195 inch ±.002.		
Dielectric core	Type A-1: Solid polyethylene. Diameter: .680 inch ±.010.		
Outer conductor	Single braid of AWG No. 30 bare copper wire. Diameter: .740 inch, maximum.		
		Alternates	
	Coverage: 97.7% nominal	97.8% nominal	97.2% nominal
	Carriers: 24	36	48
	Ends: 14	9	7
	Picks/inch: 3.1 ±10%	4.0 ±10%	5.6 ±10%

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TABLE I. Description - Continued.

Components	Construction details
Barrier tape	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape will be applied with a 50% lap, aluminum face toward the outer conductor. Diameter: .752 inch maximum.
Jacket	Cross-linked polyolefin Diameter: .870 inch ±.010.
Armor (M17/193-00002 only)	Single braid of aluminum-alloy wire Diameter: .945 inch maximum.

ENGINEERING INFORMATION:

Continuous working voltage: 8,000 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -30°C to +85°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 0.0278 ohm per 100 feet.

Elongation: 30 percent, minimum.

Engineering notes: This cable is useful in general purpose, medium low temperature applications (for M17/193-00001, see connector series "N", "C", and "SC" in accordance with MIL-C-39012; for M17/193-00002, see connector series "LC" in accordance with MIL-C-3650 and "LT" in accordance with MIL-C-26637). These cables were redesigned to meet the vertical flame test.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Out-of-roundness: Applicable.

Eccentricity: 5 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 60 pounds, minimum; 600 pounds, maximum.

Aging stability: +98°C ±2°C.

Cold bend: -30°C ±2°C.

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Dimensional stability: +85°C ±2°C.

Inner conductor from core: .200 inch, maximum.

Inner conductor from jacket: .400 inch, maximum.

Contamination: Not applicable.

Flame propagation: Applicable.

Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50

Elongation, percent of unaged minimum: 50

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness: (Type A) 80 minimum.

Weathering: Applicable.

Abrasion resistance: 75 cycles minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation, 160 percent, minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60

Elongation, percent minimum: 60

Hot oil immersion:

Tensile strength, percent minimum: 50

Elongation, percent minimum: 50

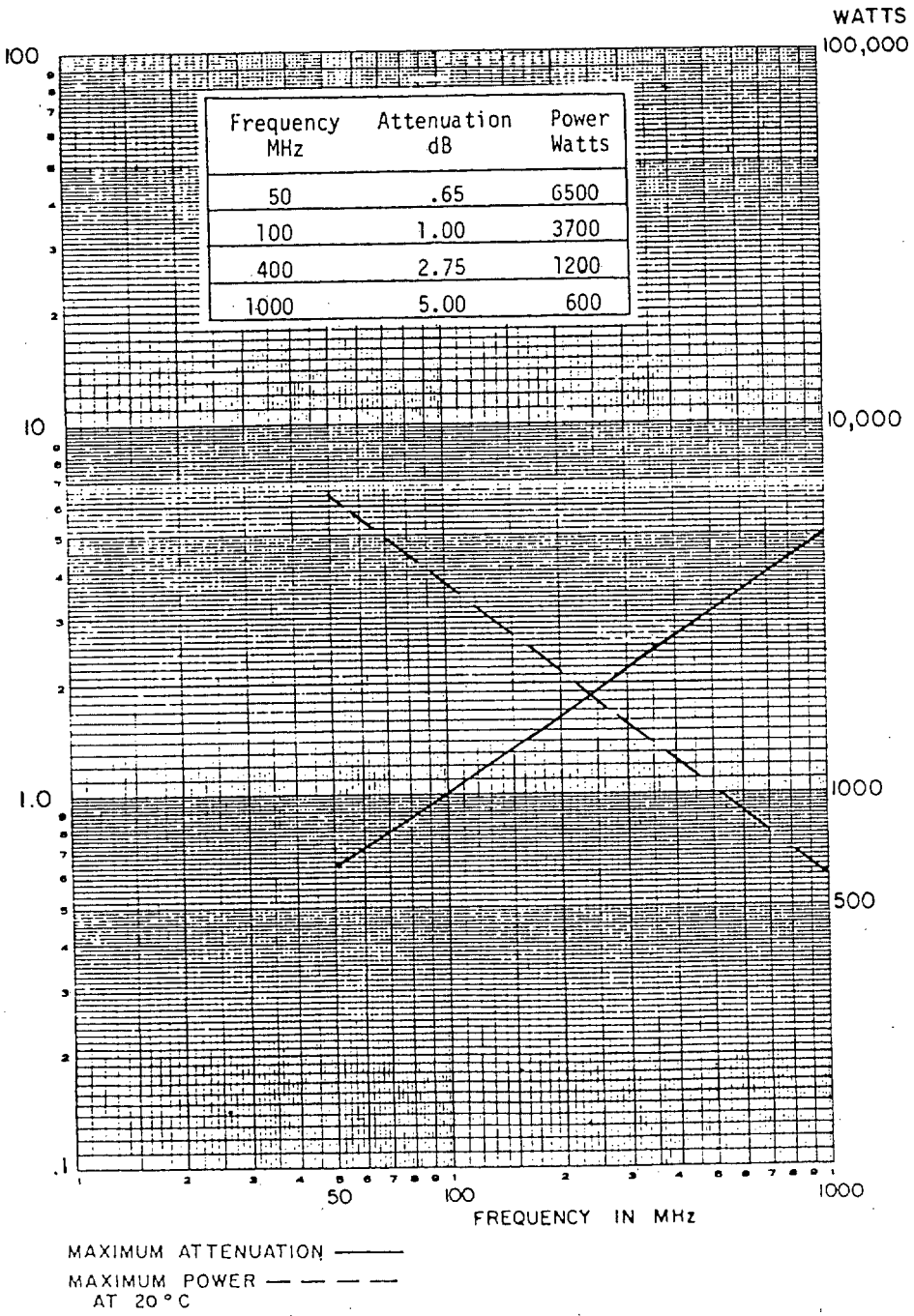
Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight:

52.1 pounds per 100 feet, maximum (M17/193-00001).

57.1 pounds per 100 feet, maximum (M17/193-00002).

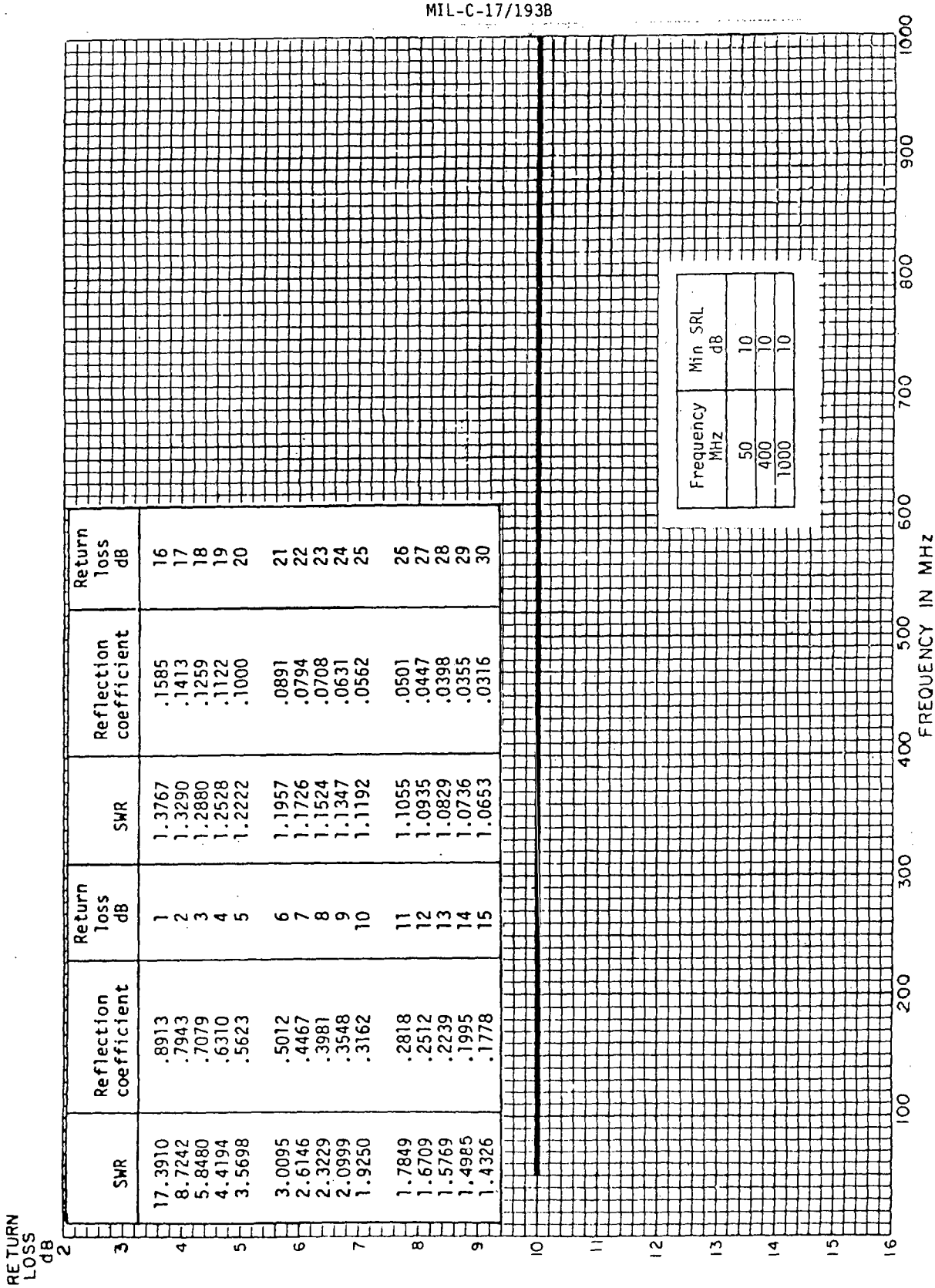
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Frequency MHz	Attenuation dB	Power Watts
50	.65	6500
100	1.00	3700
400	2.75	1200
1000	5.00	600

FIGURE 2. Power rating and attenuation.

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SWR	Reflection coefficient	Return loss dB
17.3910	.8913	1
8.7242	.7943	2
5.8480	.7079	3
4.4194	.6310	4
3.5698	.5623	5
3.0095	.5012	6
2.6146	.4467	7
2.3229	.3981	8
2.0999	.3548	9
1.9250	.3162	10
1.7849	.2818	11
1.6709	.2512	12
1.5769	.2239	13
1.4985	.1995	14
1.4326	.1778	15

SWR	Reflection coefficient	Return loss dB
1.3767	.1585	16
1.3290	.1413	17
1.2880	.1259	18
1.2528	.1122	19
1.2222	.1000	20
1.1957	.0891	21
1.1726	.0794	22
1.1524	.0708	23
1.1347	.0631	24
1.1192	.0562	25
1.1055	.0501	26
1.0935	.0447	27
1.0829	.0398	28
1.0736	.0355	29
1.0653	.0316	30

Frequency MHz	Min SRL dB
50	10
400	10
1000	10

FIGURE 3. Structural return loss.

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Electrical:

Spark test: 8,000 V rms, minimum.

Voltage withstanding: 22,000 V rms, minimum.

Corona extinction voltage: 11,000 V rms minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part or Identifying Number (PIN):

M17/193-00001 unarmored.

M17/193-00002 armored.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 80, 99
DLA - ES, IS

User activities:
Army - AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 6145-1176-14)

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